Exploratory Data Analysis

of the 1994 Adult/Census Income dataset
Part 2

The 1994 Adult Income dataset

- Used in machine learning and data analysis tasks
- It was created from the 1994 U.S. Census Bureau data
- Often used for classification and predictive modeling tasks
- Particularly for predicting whether an individual's income exceeds a certain threshold (e.g., $50,000 per year)

15 Features

Numerical Attributes

- Age
- FnlWgt
- Education-num
- Capital-Gain
- Capital-Loss
- Hours-Per-Week

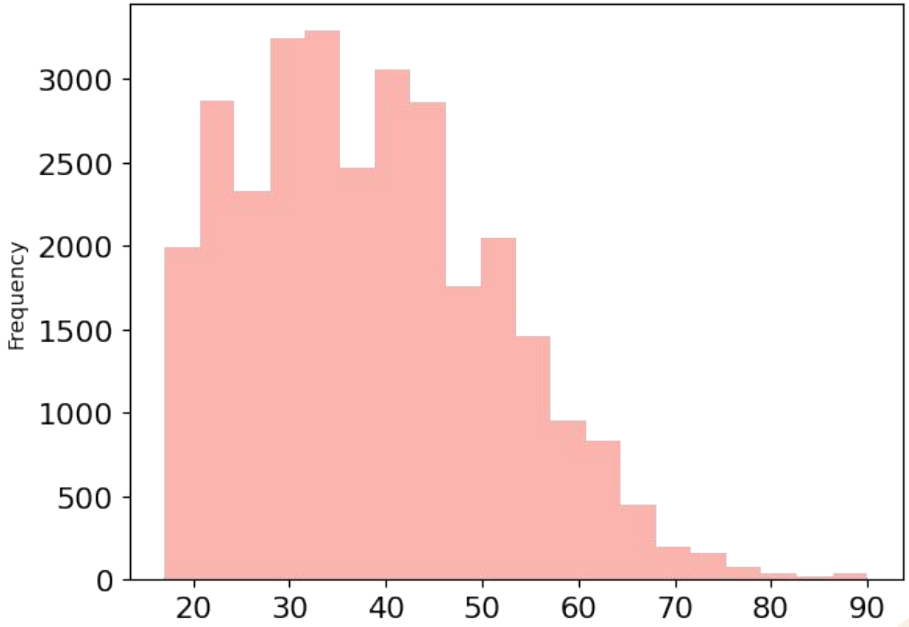
Categorical Attributes

- WorkClass
- Education
- Marital-Status
- Occupation
- Relationship
- Race
- Sex
- Native-Country
- Income

Feature: Age

- Represents the age of the individual in years
- It is a continuous numerical variable

Age



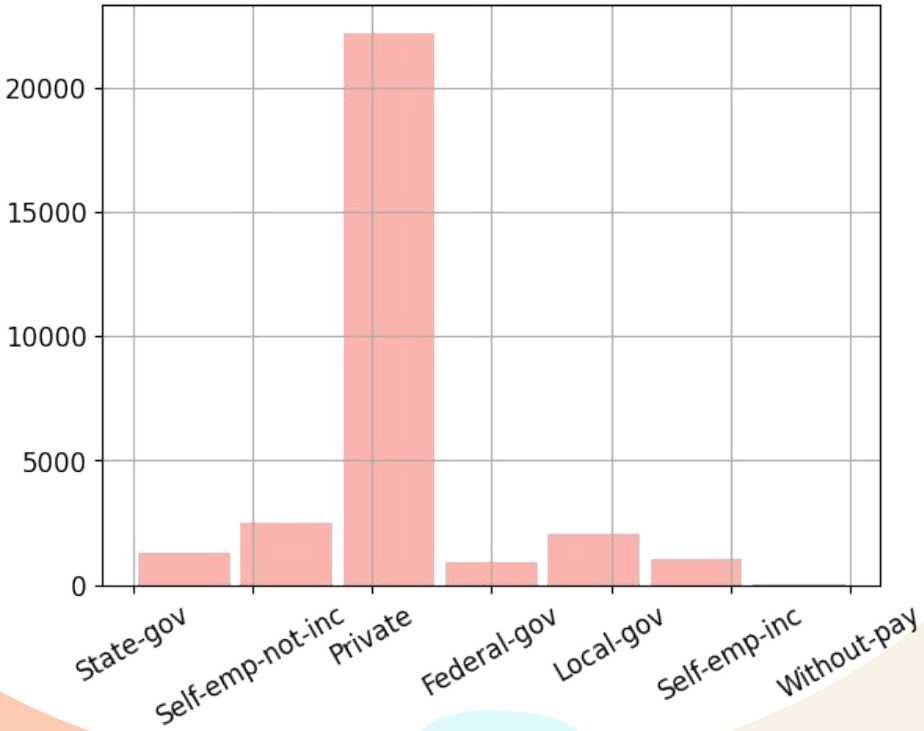
Feature: Workclass

- Describes the type of employment or workclass of the individual
- This is a categorical feature
- There are 7 categories of this feature
- It categorizes individuals based on their work situations

Categories of Workclass

- **Private**: Individuals in this category are typically employed by private companies or organizations. They receive a salary or wage from a non-governmental employer.
- **Self-emp-not-inc**: This category includes individuals who are self-employed and not incorporated. They work for themselves but do not have a formal business structure.
- **Self-emp-inc**: Similar to "Self-emp-not-inc," individuals in this category are self-employed, but they have incorporated their businesses, making them separate legal entities.
- **Federal-gov**: Individuals in this category work for the federal government of the United States. They hold government positions at the federal level.
- **Local-gov**: This category includes individuals employed by local government agencies, such as city or county governments.
- **State-gov**: Individuals in this category work for state government agencies, such as departments or agencies within a particular U.S. state.
- **Without-pay**: This category may refer to individuals who are not officially employed and may be engaged in unpaid work, volunteer activities, or other non-compensated arrangements.

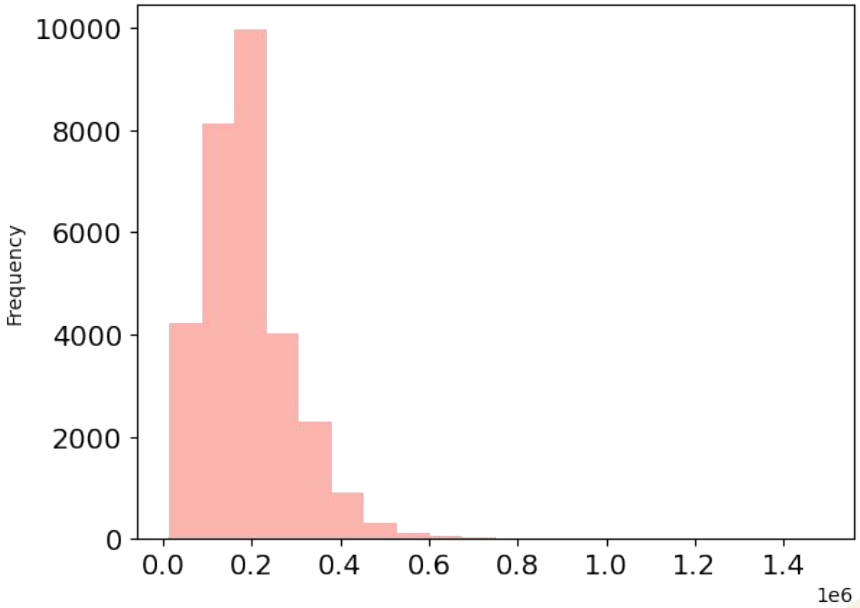
Workclass


20000
15000
10000
5000
0
State-gov
Self-emp-not-inc
Private
Federal-gov
Local-gov
Self-emp-inc
Without-pay

Feature: Final Weight

- Represents a weight assigned to each individual in the dataset
- This weight is used to extrapolate the data from the sample (the dataset) to make it more representative of the entire population
- This is a continuous numerical variable

Final Weight



Feature: Education

- Represents the highest level of education achieved by the individual
- Researchers and data analysts often use this feature to study how education level correlates with income and other socio-economic factors
- It is a critical variable for predicting income levels in the dataset, as education often plays a significant role in determining an individual's earning potential and career opportunities.
- This is a categorical feature
- There are 16 categories of this feature

Categories of Education

- **Preschool**: This category typically indicates that the individual received education at the preschool or early childhood level, which is often informal and non-standardized.
- **1st-4th**: This category represents individuals who completed up to the 4th grade in elementary school.
- **5th-6th**: Individuals in this category completed education up to the 6th grade in elementary school.
- **7th-8th**: This category includes those who completed their education up to the 8th grade in middle school or junior high school.
- **9th**: Individuals in this category completed 9th grade in high school.
- **10th**: This category represents those who completed 10th grade in high school.
- **11th**: Individuals in this category completed 11th grade in high school.
- **12th**: This category indicates that the individual completed 12th grade in high school, which is typically the final year of secondary education.
- **HS-grad (High School Graduate)**: This category signifies that the individual graduated from high school and did not pursue further formal education.

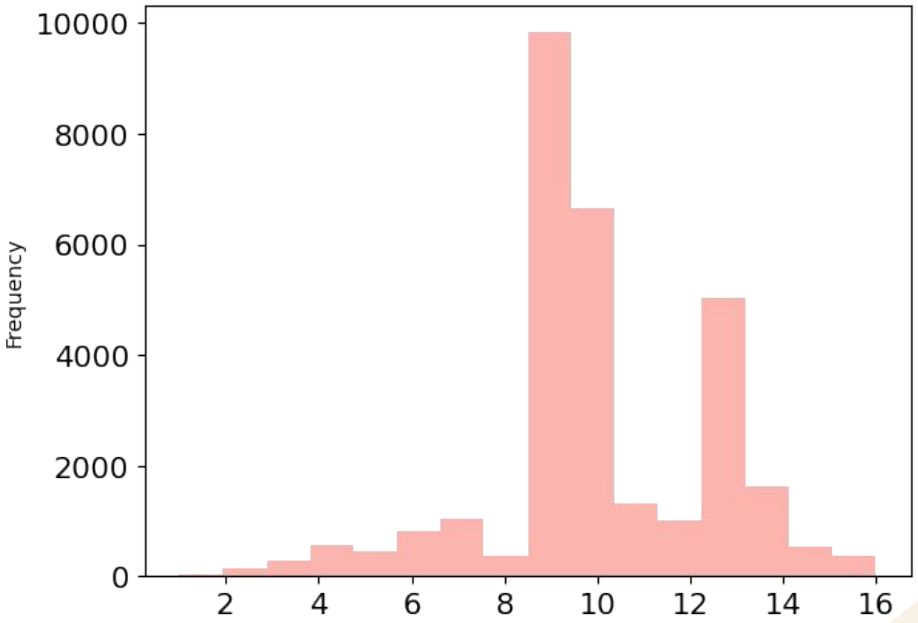
Categories of Education

- **Some-college**: Individuals in this category completed some college coursework but did not obtain a degree.
- **Assoc-acdm (Associate's Degree, Academic)**: This category represents individuals who earned an associate's degree, which is typically a two-year degree from a community college or vocational school.
- **Assoc-voc (Associate's Degree, Vocational)**: Similar to "Assoc-acdm," this category indicates individuals who earned an associate's degree but with a vocational focus.
- **Bachelors**: This category represents individuals who obtained a bachelor's degree, typically requiring four years of undergraduate study at a college or university.
- **Masters**: Individuals in this category have earned a master's degree, which typically requires additional graduate-level coursework and research beyond a bachelor's degree.
- **Doctorate (Ph.D.)**: This category indicates individuals who hold a doctoral degree (Ph.D.). Doctoral degrees often involve extensive research and the completion of a dissertation.
- **Prof-school**: This category represents individuals who attended professional school beyond a bachelor's degree. Examples include medical school, law school, or other professional programs.

Education Attribute is Ordinal

education	education-num	education	education-num
Preschool	1	HS-grad	9
1st-4th	2	Some-college	10
5th-6th	3	Assoc-voc	11
7th-8th	4	Assoc-acdm	12
9th	5	Bachelors	13
10th	6	Masters	14
11th	7	Prof-school	15
12th	8	Doctorate	16

Education



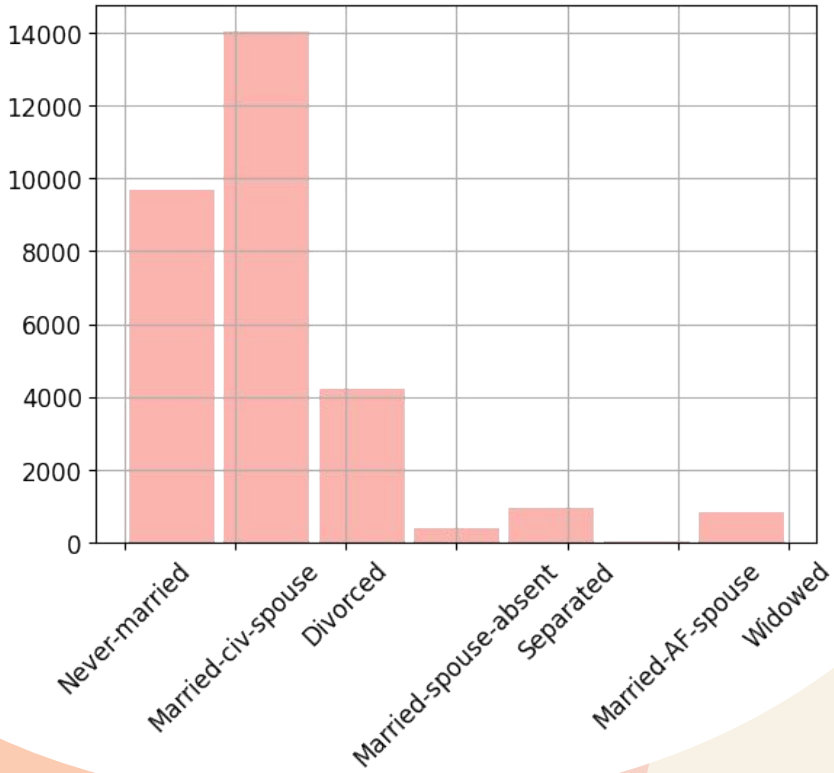
Feature: Marital Status

- Indicates the marital status of the individual
- This is a categorical feature
- There are 7 categories of this feature
- It categorizes individuals based on their current relationship or marital situation

Categories of Marital Status

- **Married-civ-spouse**: This category typically represents individuals who are legally married and living with their spouse in a civil marriage.
- **Divorced**: Individuals in this category have gone through a legal divorce process, indicating that their previous marriage has been legally dissolved.
- **Never-married**: This category includes individuals who have never been married and are currently single.
- **Separated**: Individuals in this category are legally separated from their spouses but not yet divorced. This often means they are no longer living together.
- **Widowed**: This category includes individuals who have lost their spouse due to death and are now single.
- **Married-spouse-absent**: In this category, individuals are married, but their spouse is not currently living with them, potentially due to reasons like work or travel.
- **Married-AF-spouse**: This category typically represents individuals who are married to members of the U.S. Armed Forces (e.g., Army, Navy, Air Force, or Marines).

Marital Status



Feature: Occupation

- Represents the type of occupation or job the individual is engaged in
- This is a categorical feature
- There are 14 categories of this feature
- Researchers and analysts often use this feature to understand how different occupations relate to an individual's income level and to build predictive models for income classification

Categories of Occupation

- **Tech-support**: Individuals in technical support roles typically provide assistance and support for computer-related issues or technical problems.
- **Craft-repair**: This category includes individuals who work in skilled trades and perform repair and maintenance tasks, such as carpenters, electricians, or mechanics.
- **Other-service**: This category encompasses a variety of service-oriented occupations that don't fit into more specific categories, such as private household service workers, personal care aides, or security personnel.
- **Sales**: Individuals in sales occupations are involved in selling products or services to customers. This category can include retail salespeople, sales representatives, and sales managers.
- **Exec-managerial**: This category typically includes individuals in executive and managerial roles within organizations. It encompasses roles like CEOs, managers, and executives who oversee various departments.
- **Prof-specialty**: This category represents individuals with professional and specialized occupations that often require advanced degrees or specialized training. Examples include doctors, lawyers, engineers, and scientists.

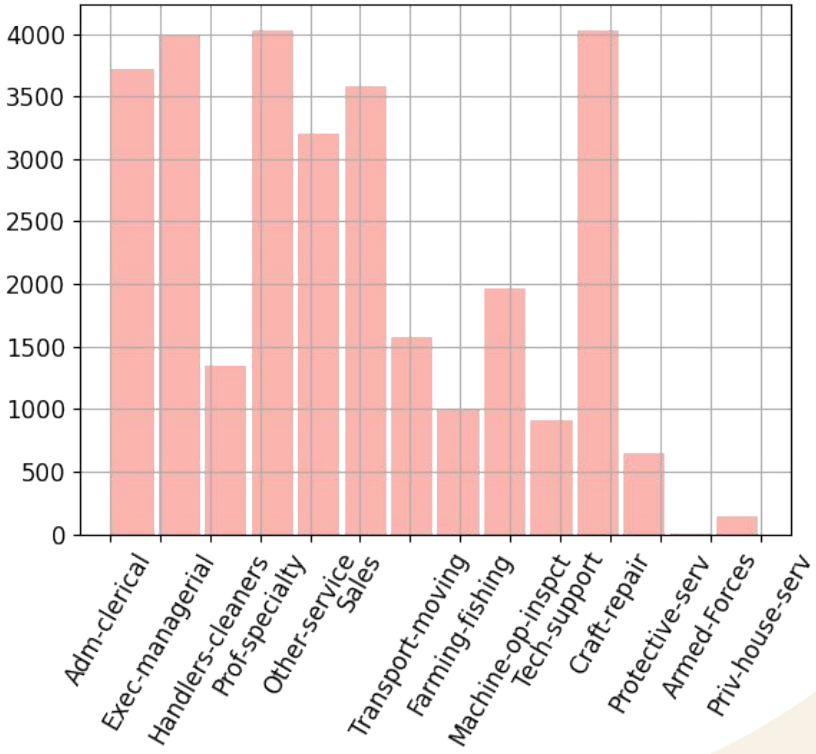
Categories of Occupation

- **Handlers-cleaners**: Individuals in this category perform manual or labor-intensive tasks related to cleaning, maintenance, or handling materials. This can include janitors, housekeepers, or agricultural workers.
- **Machine-op-inspct**: This category includes individuals who operate machines and equipment as part of their job, often in manufacturing or quality control settings.
- **Adm-clerical**: Individuals in administrative and clerical roles provide support for office and administrative functions. This includes administrative assistants, data entry clerks, and receptionists.
- **Farming-fishing**: This category includes individuals involved in agricultural and fishing occupations. They may work on farms, fisheries, or related industries.
- **Transport-moving**: Individuals in this category are typically involved in transportation and moving goods or people. It includes occupations like truck drivers, taxi drivers, and warehouse workers.
- **Priv-house-serv**: This category represents individuals working in private household service roles, such as domestic workers, housekeepers, or nannies.

Categories of Occupation

- **Protective-serv**: Occupations in this category involve providing security and protective services. This can include police officers, firefighters, and security guards.
- **Armed-Forces**: This category is for individuals who serve in the military or armed forces of a country.

Occupation


4000
3500
3000
2500
2000
1500
1000
500
0
Adm-clerical
Exec-managerial
Handlers-cleaners
Prof-specialty
Other-service
Sales
Transport-moving
Farming-fishing
Machine-op-inspct
Tech-support
Craft-repair
Protective-serv
Armed-Forces
Priv-house-serv

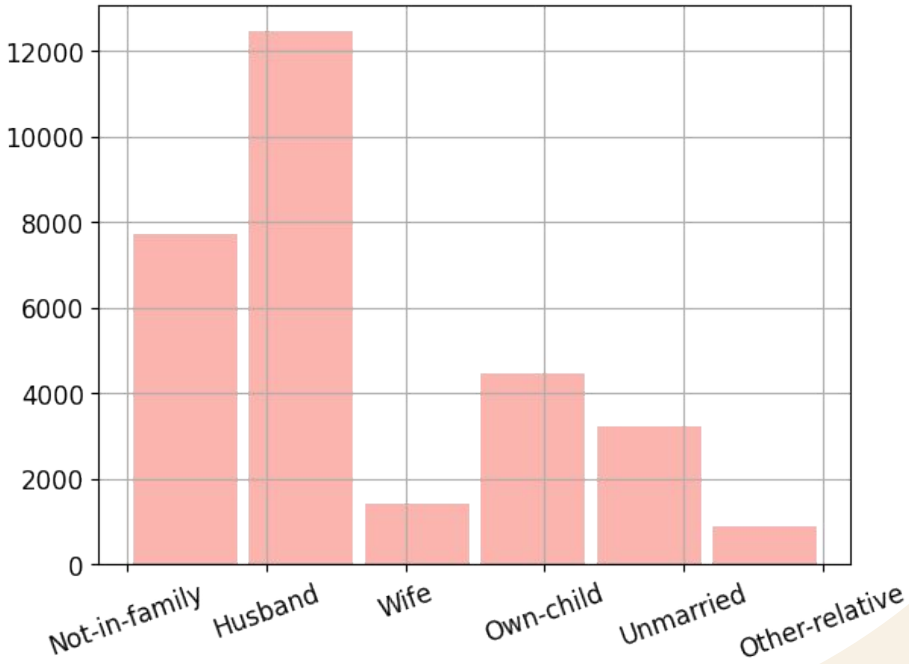
Feature: Relationship

- Describes the individual's role in the family or relationship status
- This is a categorical feature
- There are 6 categories of this feature
- It categorizes individuals based on their familial or social relationships

Categories of Relationship

- **Wife**: This category typically refers to married women who are the spouse of the household's primary earner.
- **Husband**: This category typically refers to married men who are the spouse of the household's primary earner.
- **Own-child**: Individuals in this category are the biological or adopted children of the household's primary earner.
- **Not-in-family**: This category includes individuals who do not have a familial relationship with the primary earner. They may be living alone or with unrelated individuals.
- **Unmarried**: This category typically includes individuals who are not currently married but may have other family relationships, such as siblings or parents.
- **Other-relative**: Individuals in this category are related to the household's primary earner but do not fall into the categories of spouse, child, or parent. This could include extended family members like aunts, uncles, or cousins.

Relationship



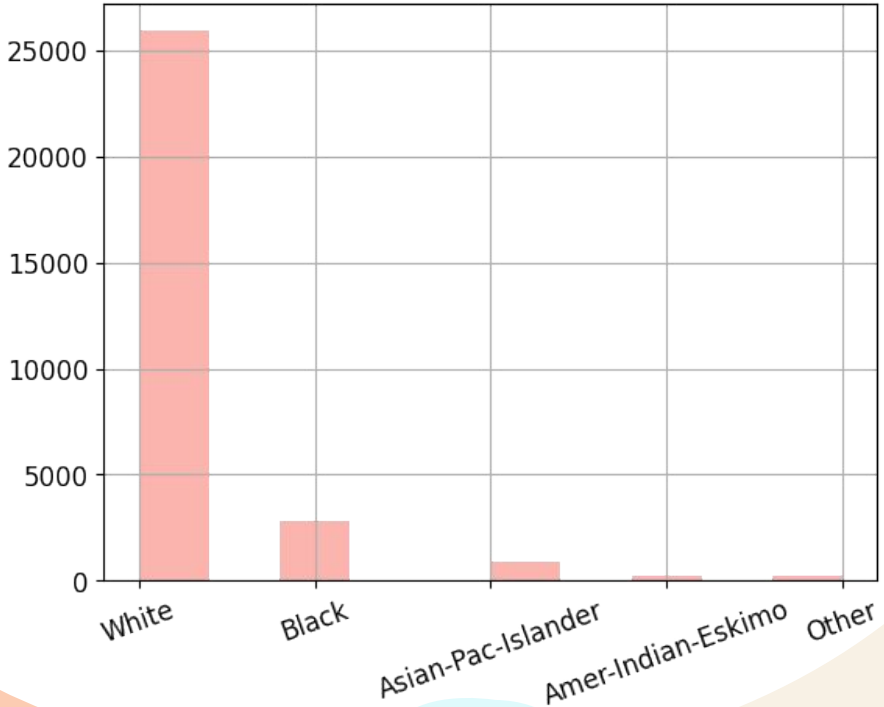
Feature: Race

- Represents the race of the individual
- This is a categorical feature
- There are 5 categories of this feature

Categories of Race

- **White**: This category typically represents individuals who are of European, Middle Eastern, or North African descent and identify as White or Caucasian.
- **Black**: This category generally includes individuals of African or African American descent.
- **Asian-Pac-Islander**: This category encompasses individuals of Asian or Pacific Islander descent. It can include people from various Asian countries, as well as those from Pacific Island nations.
- **Amer-Indian-Eskimo**: This category often includes individuals who identify as American Indian or Alaska Native. It encompasses Native Americans and indigenous peoples of Alaska.
- **Other**: The "Other" category typically includes individuals who do not fall into any of the above-mentioned racial or ethnic groups. It can encompass a wide range of diverse backgrounds.

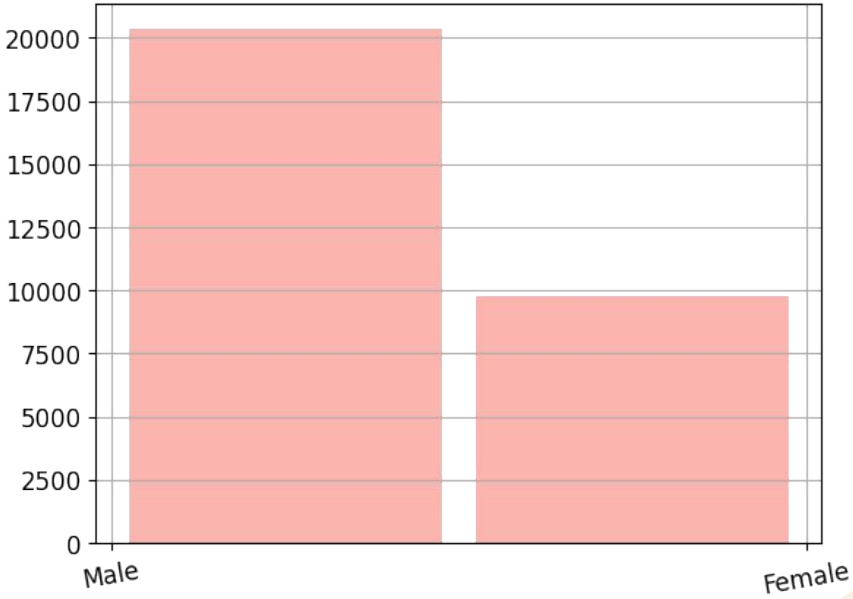
Race


25000
20000
15000
10000
5000
0
White
Black
Asian-Pac-Islander
Amer-Indian-Eskimo
Other

Feature: Sex

- Indicates the gender of the individual
- This is a binary categorical feature
- There are 2 categories of this feature
- **Male**: the individual is of the male sex.
- **Female**: the individual is of the female sex.

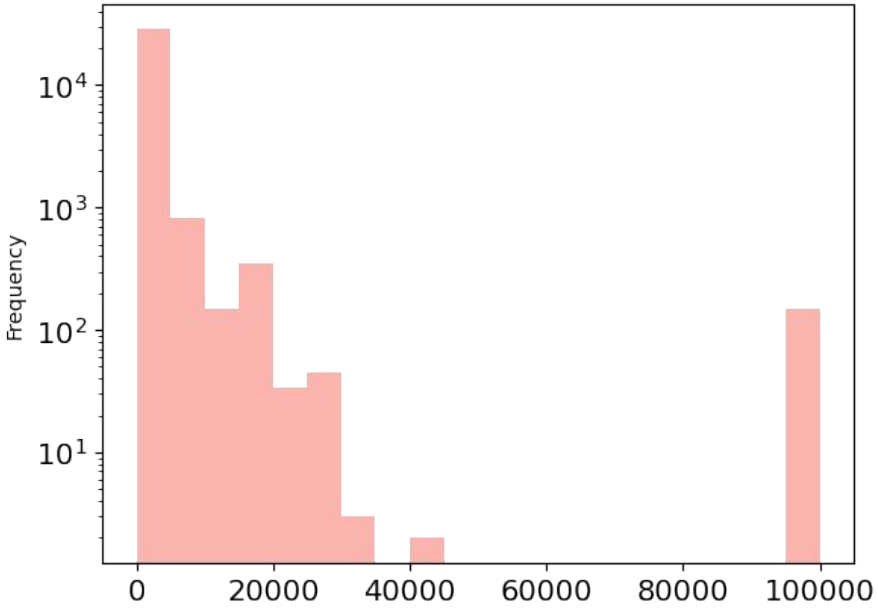
Sex



Feature: Capital Gain

- Represents the capital gains an individual has earned during the year
- This is a continuous numerical variable
- **Definition**: Capital gain is the profit realized when an asset is sold at a price higher than its original purchase price (basis). It represents the increase in the value of the asset over time.
- **Example**: If you buy a stock for $1,000 and sell it later for $1,500, you have a capital gain of $500 ($1,500 - $1,000).

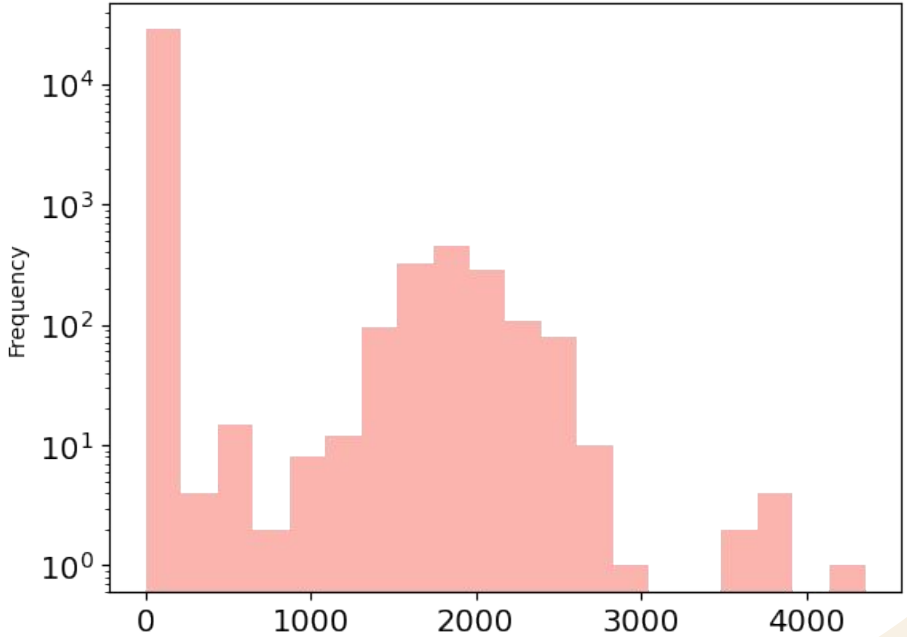
Capital Gain



Feature: Capital Loss

- Represents the capital losses an individual has incurred during the year
- This is a continuous numerical variable
- **Definition**: Capital loss is the financial loss incurred when an asset is sold for a price lower than its original purchase price (basis). It represents a decrease in the value of the asset.
- **Example**: If you buy a piece of real estate for $200,000 and sell it later for $180,000, you incur a capital loss of $20,000 ($180,000 - $200,000).

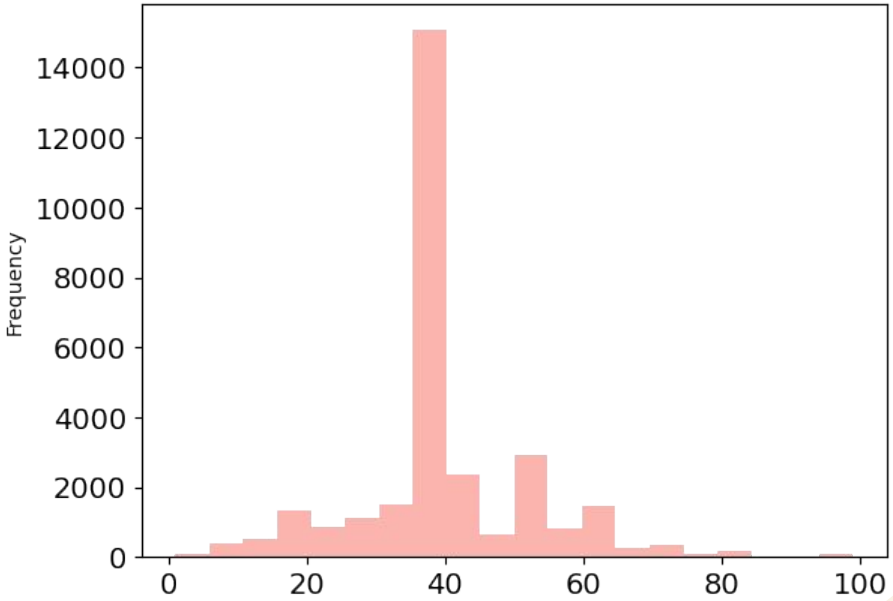
Capital Loss



Feature: Hours per Week

- Indicates the number of hours worked per week by the individual
- This is a continuous numerical variable

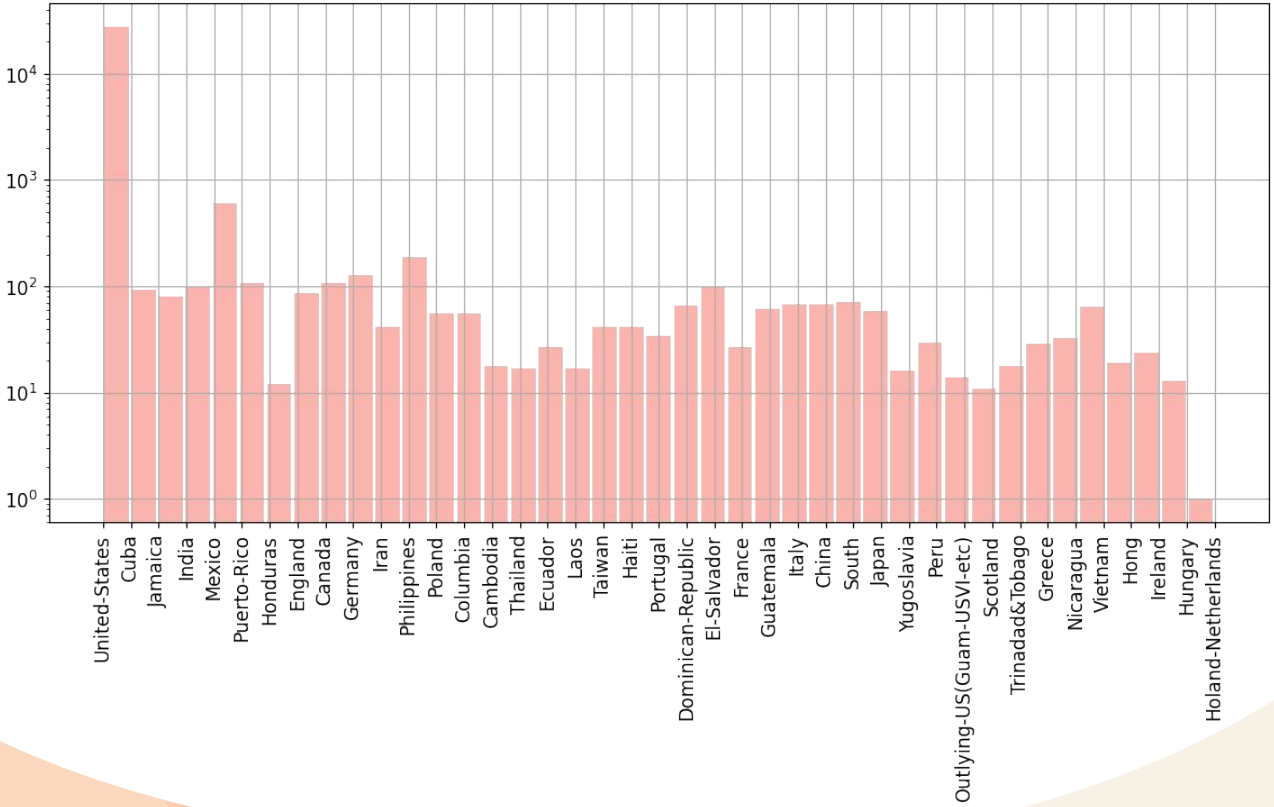
Hours per Week



Feature: Native Country

- Specifies the country of origin or nationality of the individual
- This is a categorical feature
- There are 41 categories (countries) of this feature

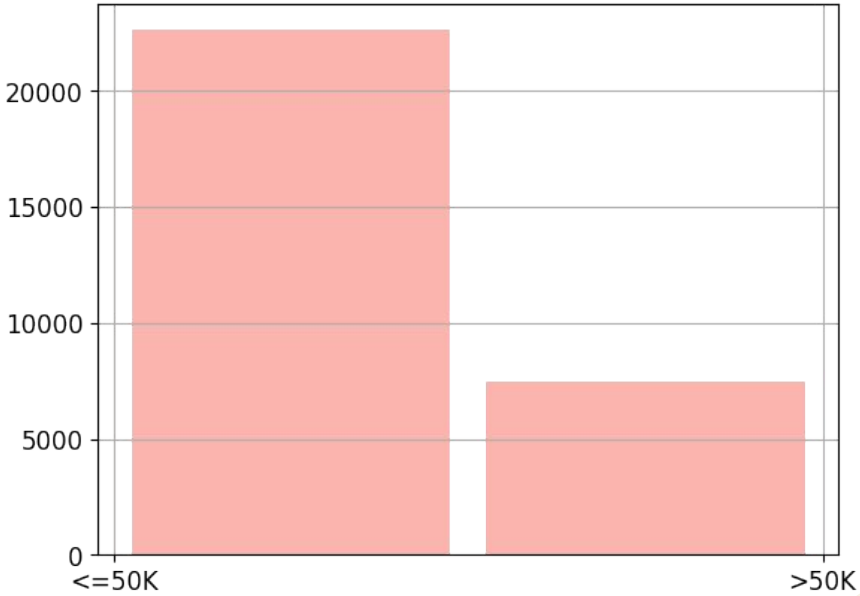
Native Country



Feature: Income

- Represents whether the individual's income exceeds $50,000 per year and is typically categorized as ">50K" (income greater than $50,000) or "<=50K" (income less than or equal to $50,000).
- This is the target variable and is categorical.

Income



20 Random Samples

age	work-class	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country	income
32	Private	94041	HS-grad	9	Married-civ-spouse	Other-service	Husband	White	Male	0	0	25	Ireland	<=50K
50	Local-gov	183390	Bachelors	13	Separated	Prof-specialty	Not-in-family	White	Male	0	0	40	United-States	>50K
40	State-gov	166327	Some-college	10	Divorced	Adm-clerical	Not-in-family	Black	Female	0	0	35	United-States	<=50K
25	Private	476334	HS-grad	9	Married-civ-spouse	Sales	Husband	White	Male	0	0	48	United-States	<=50K
43	Private	498079	Bachelors	13	Divorced	Exec-managerial	Unmarried	White	Female	0	0	40	United-States	<=50K
22	State-gov	151991	Some-college	10	Never-married	Tech-support	Own-child	White	Male	0	0	20	United-States	<=50K
34	Private	238376	1st-4th	2	Married-civ-spouse	Craft-repair	Husband	White	Male	0	0	40	Mexico	<=50K
75	NaN	222789	Bachelors	13	Widowed	NaN	Not-in-family	White	Female	0	0	6	United-States	<=50K
27	Private	133929	Bachelors	13	Never-married	Prof-specialty	Unmarried	White	Female	0	0	36	NaN	<=50K
34	Private	180284	Some-college	10	Married-civ-spouse	Handlers-cleaners	Husband	White	Male	0	0	50	United-States	<=50K
27	NaN	180553	HS-grad	9	Married-civ-spouse	NaN	Wife	White	Female	0	0	40	United-States	>50K
64	NaN	50171	HS-grad	9	Married-civ-spouse	NaN	Husband	White	Male	0	0	10	United-States	<=50K
65	Self-emp-not-inc	99359	Prof-school	15	Never-married	Prof-specialty	Not-in-family	White	Male	1086	0	60	United-States	<=50K
37	Private	143771	HS-grad	9	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	60	United-States	>50K
60	NaN	56248	HS-grad	9	Married-civ-spouse	NaN	Husband	White	Male	0	1485	70	United-States	>50K
44	Private	275094	1st-4th	2	Never-married	Other-service	Own-child	White	Male	0	0	10	United-States	<=50K
31	State-gov	161631	Some-college	10	Divorced	Adm-clerical	Unmarried	White	Female	0	0	40	United-States	<=50K
38	Private	99138	Bachelors	13	Never-married	Exec-managerial	Not-in-family	White	Female	0	0	38	United-States	<=50K
56	Private	103948	11th	7	Married-civ-spouse	Transport-moving	Husband	White	Male	0	0	40	United-States	<=50K
49	Self-emp-not-inc	263568	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	60	United-States	>50K

1,843

occupation missing values

1,836

work-class missing values

583

native-country missing values

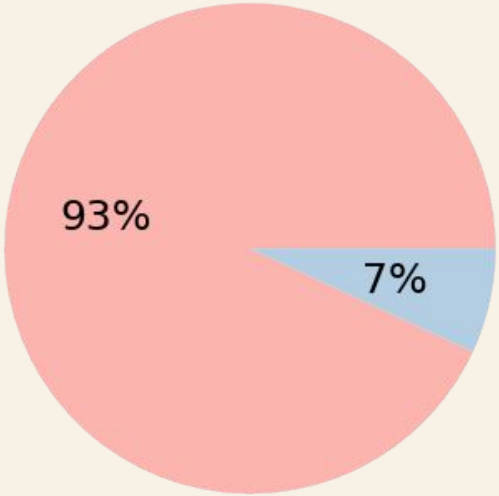
No Missing Values vs. Missing Values

No missing

93% of the rows have no missing values

Missing

7% of the rows have some missing values



From now on our Adult dataset will only contains rows with **no missing values** and **no duplications**.

Descriptive Statistics

Attr.	Mean	Min	Max	Std.	75%
age	38.44	17.00	90.00	13.13	47.00
education-num	10.12	1.00	16.00	2.55	13.00
capital-gain	1092.84	0.00	99999.00	7409.11	0.00
hours-per-week	40.93	1.00	99.00	11.98	45.00

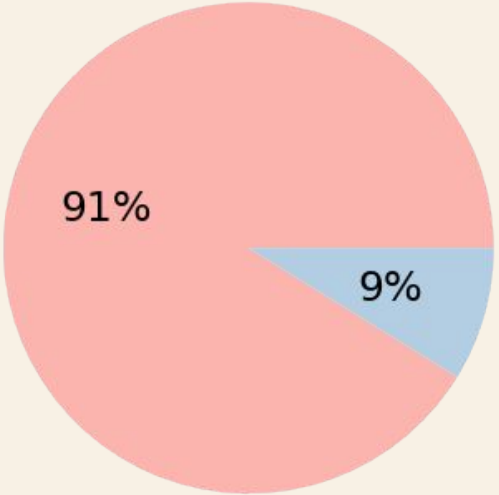
United States vs. Other Countries

United States

91% of adults in the dataset are native to US

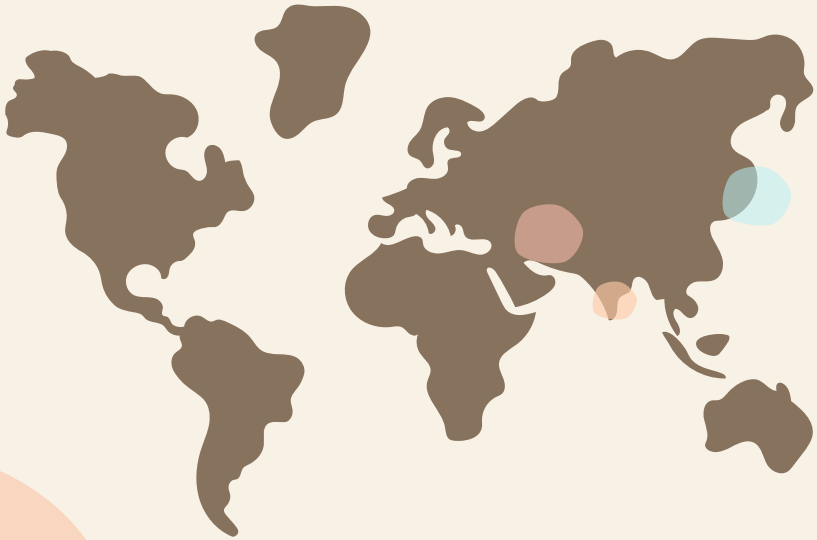
Other Countries

9% of adults in the dataset are native to some other countries



Only 17 or 0.056% of adults in the dataset are native to Thailand.

Top 3 Countries with Highest Average Education



13.48

Taiwan

75% of people with bachelor or higher.

12.45

Iran

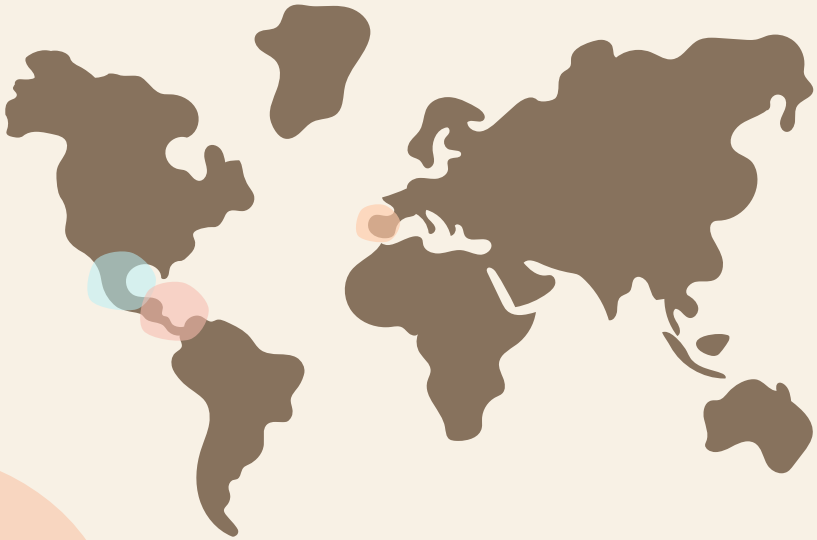
75% of people with Some-college or higher.

12.43

India

75% of people with Some-college or higher.

Top 3 Countries with Lowest Average Education



6.15

Mexico

75% of people with HS-grad or lower.

6.20

Guatemala

75% of people with HS-grad or lower.

6.85

Portugal

75% of people with HS-grad or lower.

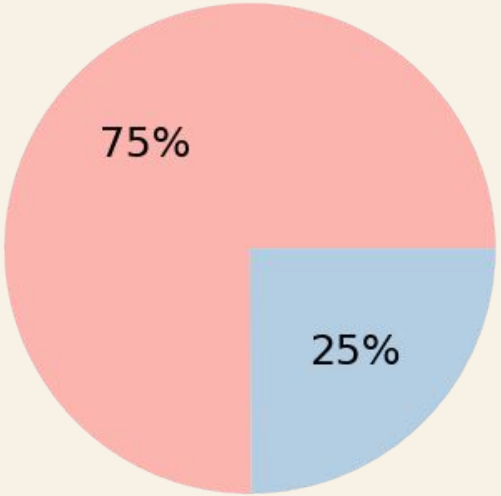
Yearly Income

<= 50K

75% of adults have yearly income less than or equal to 50,000 USD

> 50K

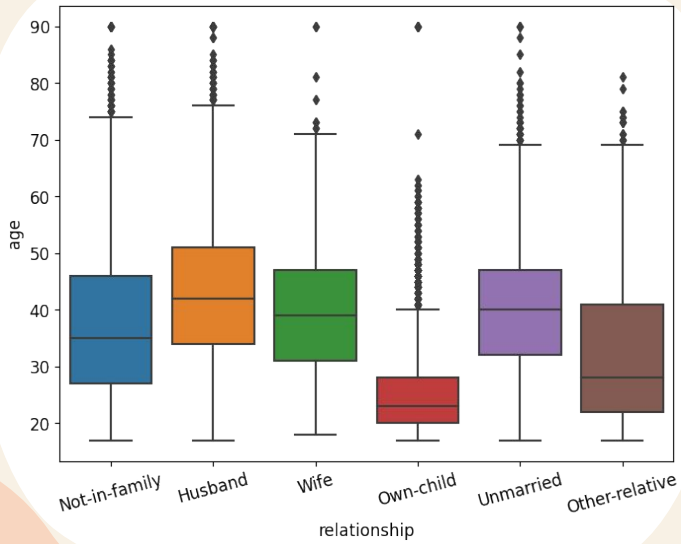
25% of adults have yearly income more than 50,000 USD



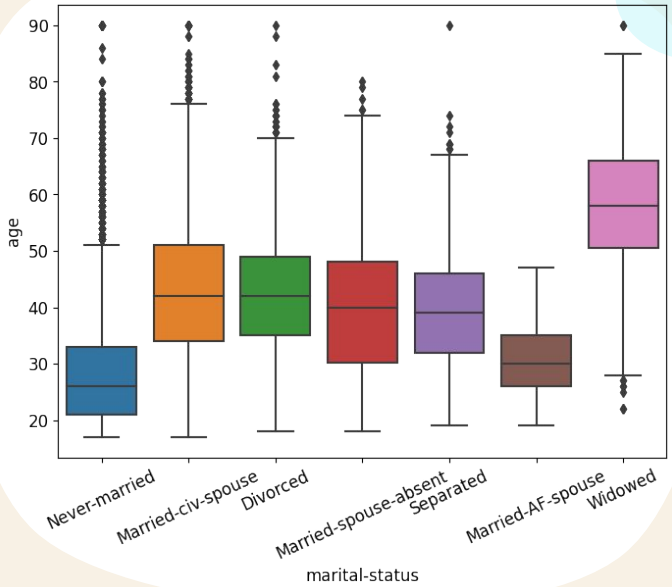
Due to the fact that the original dataset was from the US, It's safe to assume that the monetary unit was USD.

Box Plot

Age & Relationship

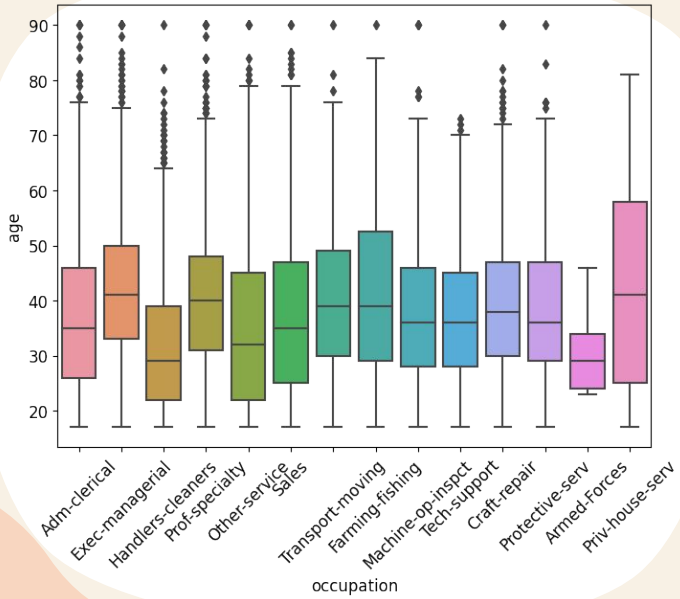


Age & Marital-Status

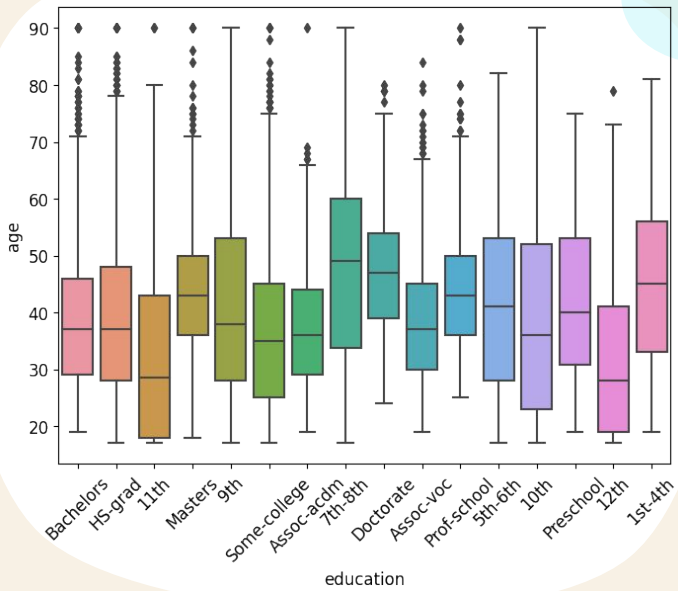


Box Plot

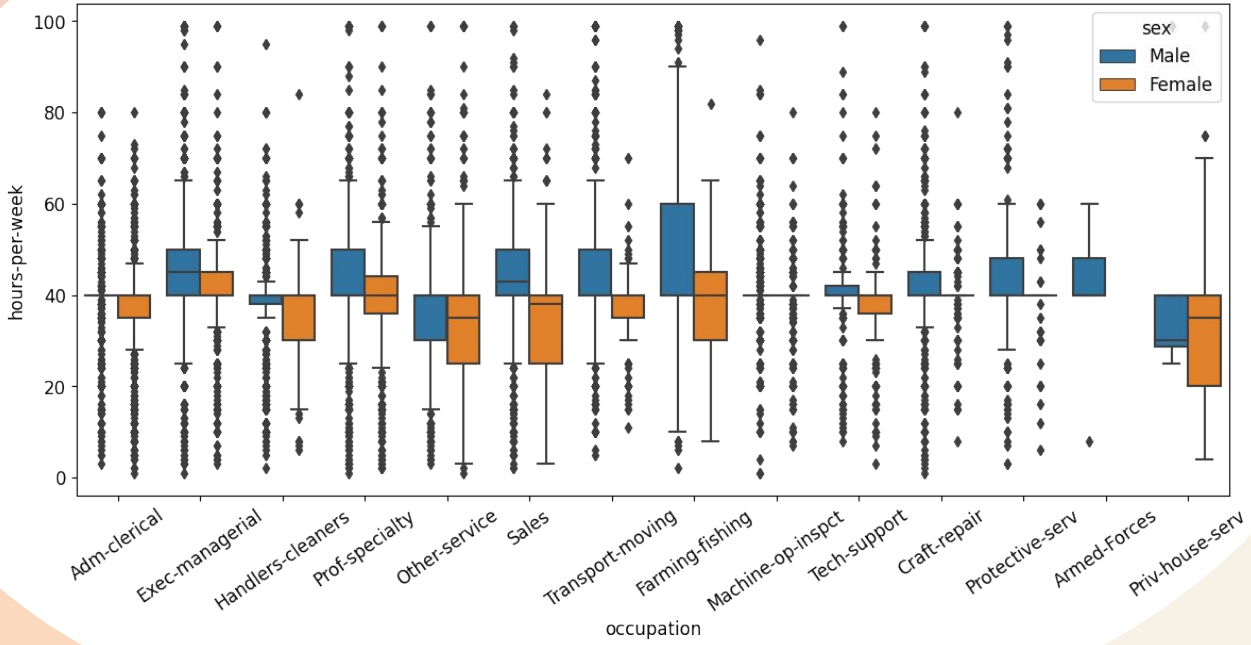
Age & Occupation



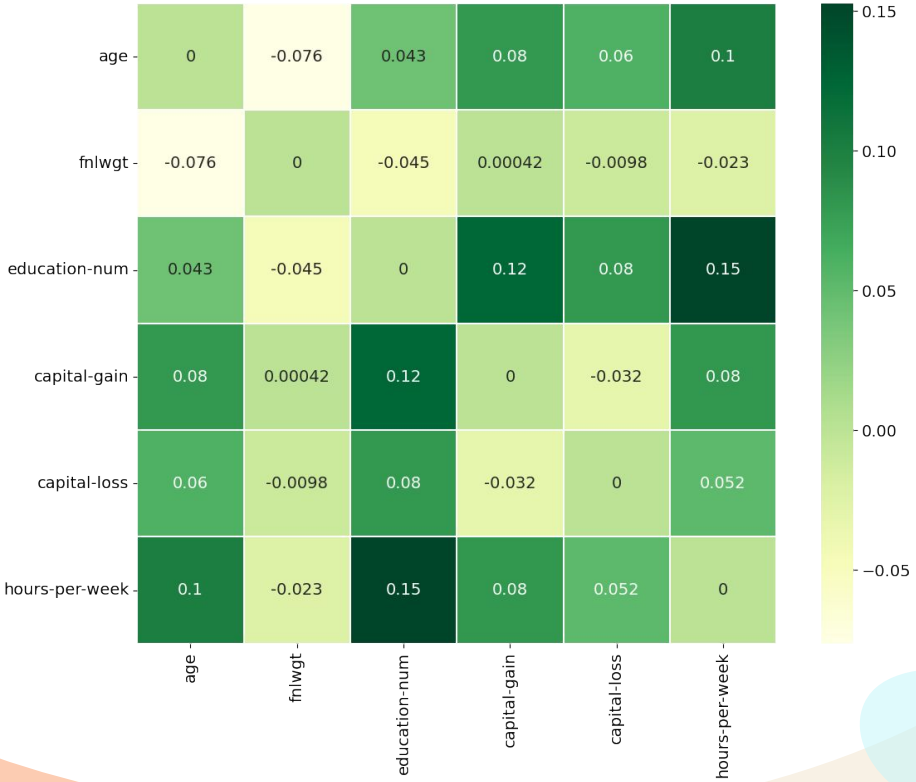
Age & Education



Hours-per-week & Occupation



Pearson Correlation of Numerical Features



Pearson Correlation Formula

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

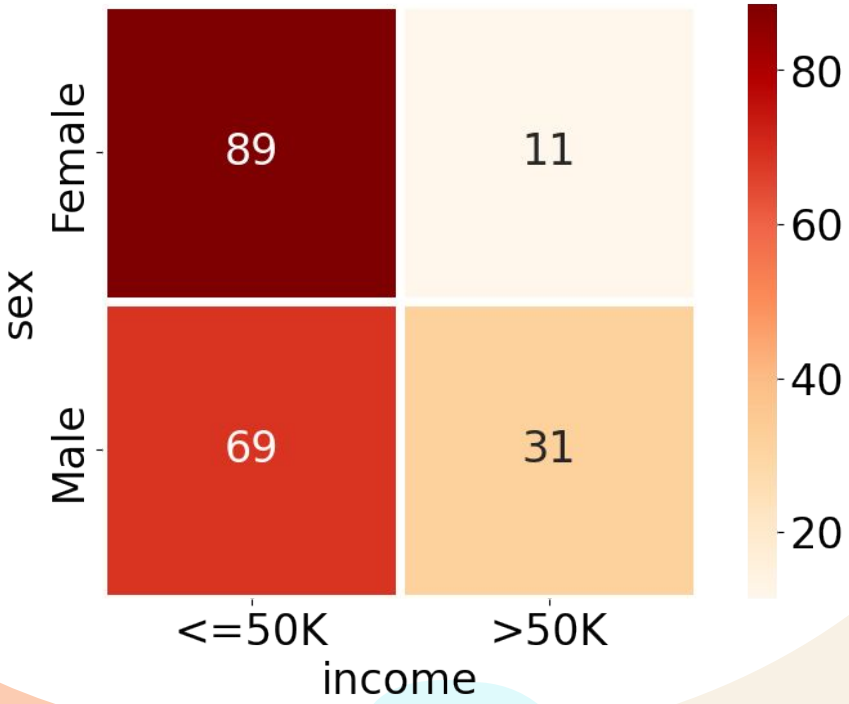
x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

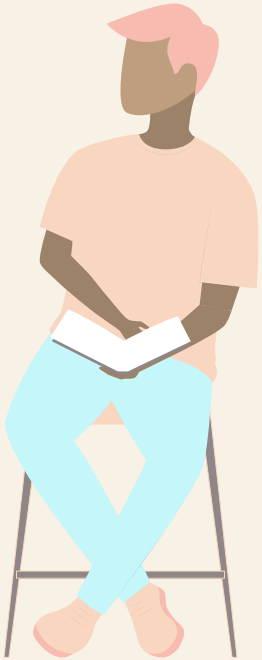
$\bar{y}$ = mean of the values of the y-variable

Sex-Income Contingency Table



Occupation-Education Contingency Table

1: Preschool
2: 1st-4th
3: 5th-6th
4: 7th-8th
5: 9th
6: 10th
7: 11th
8: 12th
9: HS-grad
10: Some-college
11: Assoc-voc
12: Assoc-acdm
13: Bachelors
14: Masters
15: Prof-school
16: Doctorate

occupation \ education-num	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Adm-clerical	0.054	0	0.16	0.3	0.38	1	1.8	0.97	36	34	4.4	5.1	13	1.8	0.22	0.11
Armed-Forces	0	0	0	0	0	0	0	11	44	22	0	0	11	11	0	0
Craft-repair	0.099	0.57	1	2.7	2.3	4	4.3	1.4	47	21	6.2	2.8	5.4	0.52	0.17	0.05
Exec-managerial	0	0.075	0.025	0.45	0.3	0.6	0.83	0.33	20	22	3.8	3.6	33	12	1.2	1.4
Farming-fishing	0.81	1.8	3.6	7	2.8	4.4	3.7	1.6	41	18	5.3	1.4	7.8	1	0.41	0.1
Handlers-cleaners	0.15	1.2	2.8	3.4	3.6	5.3	9	2.7	45	19	2.1	1.6	3.6	0.37	0	0
Machine-op-inspct	0.56	1.2	2.5	4.7	3.8	5.1	5	1.7	51	16	3.2	1.7	3	0.41	0	0.051
Other-service	0.44	1.2	1.9	2.8	3.1	6	7.4	2.6	39	24	3.4	2.4	5.3	0.56	0.12	0
Priv-house-serv	1.4	6.4	9.9	5.7	7.1	4.3	9.9	2.8	33	11	2.8	1.4	4.3	0	0	0
Prof-specialty	0.025	0.099	0.025	0.2	0.074	0.22	0.5	0.22	5.7	10	4.2	3.4	36	20	11	7.5
Protective-serv	0	0.16	0.16	1.4	0.62	0.93	1.1	0.78	33	31	7.5	5.3	16	2.3	0.16	0
Sales	0	0.2	0.31	0.81	0.89	2.2	3.9	1.2	29	28	2.9	4	22	3.6	0.47	0.22
Tech-support	0	0	0.11	0.55	0.22	0.33	0.66	0.33	17	30	14	7.9	24	3.7	0.77	0.22
Transport-moving	0	0.45	1.7	3.8	2.2	5.3	5.9	2.4	52	18	2.5	1.7	3.5	0.57	0.19	0.064

Thanks For Listening